

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7.	(a)			(i) (graphite) anode (1) (ii) Molten aluminium (1) (iii) (graphite) cathode (1)	3			3		
	(b)			2 Al₂O₃ → 4 Al (1) + 3 O₂ (1) one mark for balancing (1) formulae must be correct to obtain balancing mark		3		3		
	(c)			Any 3 × (1) from: At cathode/negative electrode – aluminium <u>ions</u> gain (three) electron(s) (1) (Is reduced) and forms aluminium <u>atoms</u> (1) At anode / positive electrode – oxide (ions) lose electrons / oxidized (1) Forms oxygen (molecules) (1) Allow for (1) Aluminium ions gain electrons and oxide (ions) lose electrons	3			3		
	(d)	(i)		Any 2 × (1) from: Less cost of transport of Al ore to the UK (1) Lower cost of labour (1) no {mining/extraction} aluminium (ore) is required (1) less electrical energy needed to recycle / lower energy cost (1) do not accept 'less cost' unless qualified	2			2		

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		(ii)		Less waste in landfill (1) Less habitats destroyed / less land used (1) OR Less energy used for recycling (1) So less impact on global warming / climate change / greenhouse effect / carbon footprint (1)	2			2		
				Question 7 total	10	3	0	13	0	0